

NG PULSE

Rail-Signature Workload Integrity Monitor

EEL 4914 Senior Design I
Divide and Conquer Proposal Draft

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Executive Summary

NG PULSE is a hardware-security monitoring project intended to detect physical attacks against a protected processor by observing its supply-rail current behavior. The proposed monitor will attach to any printed circuit board (PCB), acquire current signatures from the host processor, compare measured behavior against a known-good reference or golden profile, and generate an alert when an abnormal condition is detected. An advanced implementation may also initiate a protective system shutdown after an attack is identified.

The planned architecture combines a current-sense front end, a 12-bit analog-to-digital converter (ADC), windowed feature extraction, and threshold or classification logic. The project will first establish reference current waveforms for selected RISC-V processor functions or workloads. The detection logic will then be implemented in Verilog and verified before physical prototyping. The prototype is planned as a PCB containing the current-sense design and MCU for processing, to monitor a ESP32-C (board with a RISC-V processor) during normal function.

The project is funded by Northrop Grumman with a maximum budget of \$1,000. Key performance targets include 100% coverage of the reference waveforms defined by the project test plan, no more than a 5% reduction in RISC-V computing efficiency, no more than 1 second of attack-detection latency, at least a 95% attack-detection rate, no more than a 5% false-positive rate, 12-bit ADC resolution, and operation from either a 3.3 V or 5 V input supply.

1. Motivation and Background

Modern computing systems depend on integrated circuits that may be exposed not only to software-based threats but also to physical attacks that interfere directly with hardware operation. NG PULSE is motivated by the need for an additional layer of hardware-level security that can identify abnormal electrical behavior in a protected processor. A representative threat identified for this project is abnormal or high-current injection intended to disturb normal chip operation.

The central concept is to monitor the processor's supply-rail current signature. Under normal operation, processor activity produces characteristic electrical behavior. By establishing a golden representation of that behavior and continuously comparing later measurements against it, NG PULSE is intended to identify departures that may indicate an attack. Because the monitor does not need to sit in the processor's normal control path, the design goal is to provide security monitoring while limiting the impact on host computing performance.

The project also provides a mixed-signal design flow spanning analog sensing, analog-to-digital conversion, digital signal processing, classification or threshold logic, Verilog/RTL development, CAD-based schematic integration, and eventual

physical prototyping. The final design is intended to be suitable for attachment to an existing PCB rather than requiring a complete redesign of the protected system.

2. Goals and Objectives

2.1 Chief Goal

The chief goal of NG PULSE is to design a working hardware-security monitor that can be attached to an existing PCB, detect a defined physical attack by recognizing abnormal supply-rail current behavior, and produce an alert. Where the final design and target platform permit, the monitor will also support a protective response capable of shutting down the system after an attack is detected.

2.2 Development Objectives

- Model normal operation of the ESP32-C target using appropriate software and laboratory resources.
- Obtain the golden current waveforms that represent normal processor functions or workloads defined by the project test plan.
- Select the required hardware, including the ADC, current sensor, comparator, voltage-regulation components if necessary, and supporting circuitry.
- Develop the NG PULSE digital monitoring and detection logic in Verilog.
- Develop the ASIC/system schematic and verify electrical connections using appropriate CAD software.
- Complete virtual testing and verification before beginning physical prototyping.
- Build and integrate a PCB prototype (of the current monitor only) during Senior Design II and evaluate it against the project's engineering specifications.

2.3 Basic, Advanced, and Stretch Goals

Basic Level

Goal:

Produce a functional monitor design that detects abnormal current behavior and asserts an alert.

Objectives:

- Acquire the golden current profile for normal processor operation.

- Select the necessary hardware components.
- Develop the system schematic.
- Develop the Verilog-based detector.
- Perform virtual verification of the design.
- Construct a physical PCB with attachment/integration capability.

Stretch Level

Goal:

Add an automated protective response and further integrate and optimize the mixed-signal design.

Objectives:

- Implement system shutdown capability in response to a detected abnormal condition.
- Improve overall system integration.
- Perform additional design optimization.
- Potentially incorporate Bluetooth connectivity with a mobile phone application for monitoring or system interaction.

3. System Features and Functional Description

NG PULSE will observe the supply-rail current of the protected processor and convert the sensed analog behavior into digital data. The digital subsystem will process the measurements over defined time windows and compare extracted information against the golden reference profile. When the measured behavior satisfies the project's criteria for an anomalous or attack condition, an attack flag will be asserted.

- Supply-rail current sensing.
- 12-bit analog-to-digital conversion.
- Acquisition and storage/use of golden reference waveforms.
- Windowed feature extraction or equivalent waveform-processing method.
- Threshold-based or classifier-based anomaly decision logic.
- Attack flag generation within the specified latency.
- Alert/status reporting; PMBus or I²C is identified in the project concept, with the final interface to be confirmed.
- Optional system shutdown response for the stretch implementation.
- Operation from either a 3.3 V input supply.

4. Existing Products and Prior Related Work

The project concept overlaps with several established technology categories: current-sense amplifiers, ADC-based power monitors, threshold/comparator protection circuits, processor power-signature analysis, and physical fault-injection security research. However, the information currently provided for this report does not establish a specific commercial product as a direct equivalent to NG PULSE.

4.1 Texas Instruments INA233

The INA233 from Texas Instruments is a ten pin Very Thin Shrink Small Outline Package (VSSOP). This device acts as a digital current-sense amplifier that is compatible with several interfaces and can be placed directly onto a programmable circuit board. The INA233 takes measurements of the current, voltage, and power and can report these values to enable accurate decision making in controlled systems. Another similar feature is the ability to program out-of-range limits that can issue alerts that are dependent on measured voltage and current values. With an integrated ADC, the INA233's averaging modes can also be adjusted.

4.2 Analog Devices LTC2945

The LTC2945 from Analog Devices is a twelve-pin Mini Small Outline Package (MSOP) that can be placed directly onto a PCB. This device can measure current, voltage, and power while storing values and sending alerts. The measurements taken can be adjusted to be either singular or continuous. Like the INA233, this device will also send an alert when measured values are outside of the acceptable parameters. This device's integrated shutdown mode also reduces power consumption to 20 microamps.

4.3 Analog Devices ADM1293

Analog Devices' second comparison is the ADM1293. This is a fourteen-pin Thin Shrink Small Outline Package (TSSOP). This device also measures the voltage, current, and power of a monitored rail. An integrated 12-bit ADC measures the voltage to determine the load current, and is capable of calculating the current power with these values. Like the LTC2945, the sampling can be adjusted to be either singular or continuous with the ability to reduce sensing noise. By storing measured values of normal behavior, the ADM1293 can use programmable thresholds to detect abnormal conditions such as overcurrent or overvoltage. Any observed abnormal behaviors can then trigger an alert.

5. Engineering Specifications

The following specifications define the current quantitative targets for the NG PULSE system. The final test plan should define the exact workloads, attack cases, measurement equipment, and pass/fail procedures used to demonstrate each requirement.

Table 5.1: Engineering Specifications

Category	Requirement	Verification Method
Reference waveform coverage	For all (4) selected processes, collect and average 1024 current samples for golden profile	Golden-profile acquisition records
RISC-V efficiency impact	Zero tolerance for reduction in computing efficiency with NG PULSE attached. (0% efficiency loss)	Benchmark with/without monitor
Attack detection latency	Maximum 1 s from detectable attack/anomaly to asserted flag	Timed attack/flag measurement
Attack detection rate	Minimum 95%	Repeated defined attack trials
False-positive detection rate	Maximum 5%	Repeated normal-workload trials
Current sensing range	500mA to 20A	Thermal imaging camera under high load and minimum current detection within specified accuracy.
Signal to noise ratio	Minimum 20dB	Golden profile comparison to actual waveform
F1 score	Minimum 0.85	Repeated and defined attack trials and normal operation monitoring
Long time overcurrent shutoff	For a surge of 1.05-1.3pu, shutoff occurs within 20s	Repeated and defined attack trials
Short time overcurrent shutoff	For a surge of 1.3-2pu, shutoff occurs within 5s	Repeated and defined attack trials

Instantaneous overcurrent shutoff	For a surge of 2pu or greater, shutoff occurs within 0.1s	Repeated and defined attack trials
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6. Proposed Hardware Architecture

At a high level, the protected processor is powered through a rail that is observed by the NG PULSE sensing circuitry. The current-sense front end produces a measurable analog representation of the rail current. A 12-bit ADC digitizes this signal. Digital processing then extracts waveform information and compares the measured behavior against the golden profile. The decision engine asserts an attack flag when the anomaly criteria are satisfied. The flag may be reported externally and, in the advanced implementation, may initiate a shutdown response.

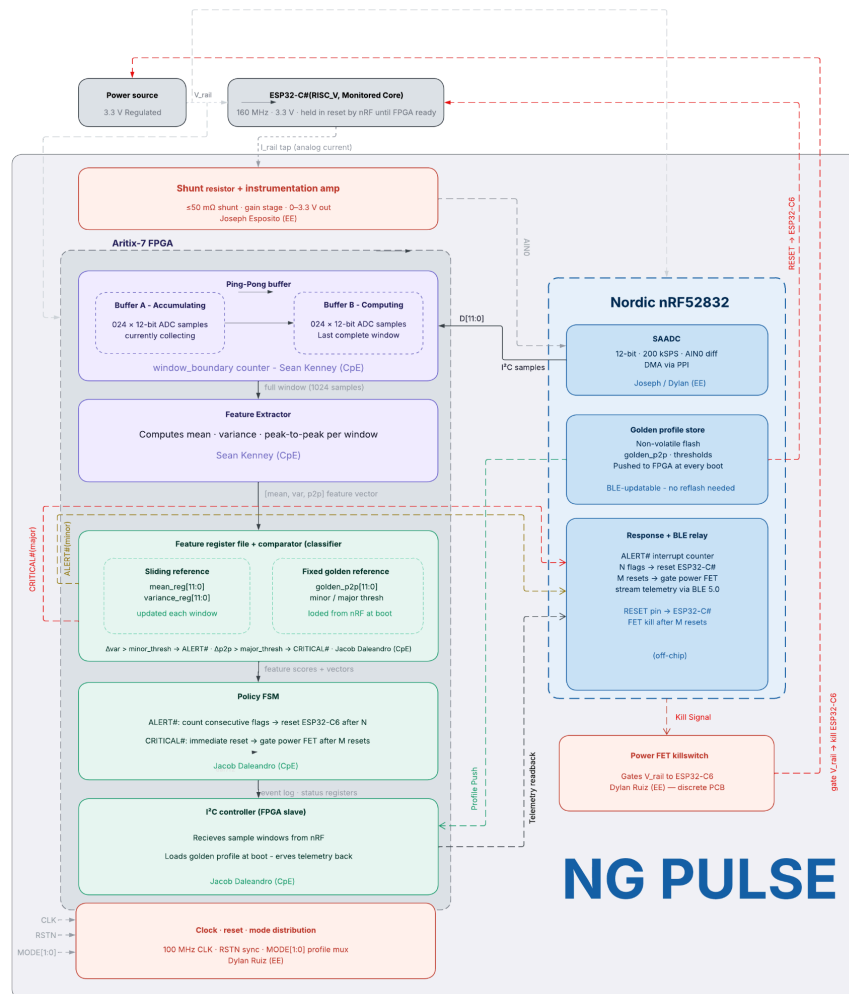


Figure 6.1: Hardware Block Diagram

6.1 Preliminary Component Categories

- Shunt resistor
- Op-amp
- MCU: Nordic
- FPGA/development platform for prototype
- Target RISC-V processor/platform: ESP32-C
- Communication/interface components:TBD
- Protection/shutdown circuitry: MOSFET

7. Digital Logic and Software/RTL Flow

The NG PULSE digital subsystem will be developed in Verilog. The intended processing flow begins with digitized current samples from the ADC. Samples will be organized into appropriate analysis windows, processed to obtain the features or values required by the detector, and evaluated against the golden profile or configured decision thresholds. When the decision logic identifies an attack condition, it will assert the attack flag and provide the required reporting or protection response.

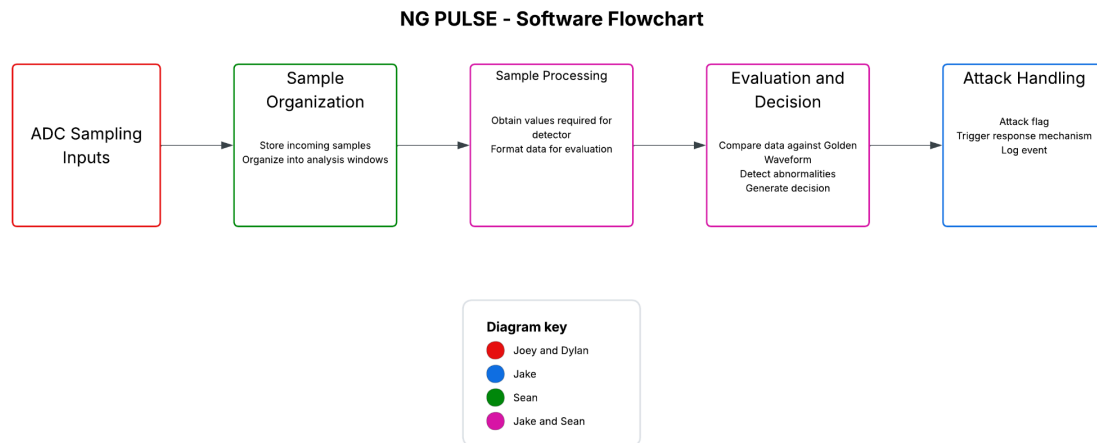


Figure 7.1: Software Flowchart

8. Prototype Plan

The initial physical detector will be implemented on a PCB using commercial off-the-shelf components. The PCB will contain the sensing, controls, and clock signals, as well as pin headers for the ESP32-C, FPGA, and Nordic board, and supporting power circuitry required to evaluate the NG PULSE concept.

The preferred prototype will connect the detector to an FPGA available in the Senior Design laboratory so that the Verilog logic can be implemented, tested,

and modified rapidly. The team will connect the prototype to a RISC-V processor board (ESP32-C) in the DRACO laboratory. This would allow the monitor to acquire and evaluate current signatures from an actual RISC-V target.

Prototype testing will verify current acquisition, golden-profile comparison, attack flag generation, detection latency, detection rate, false-positive rate, and all other specifications listed in table 5.1.

9. House of Quality

A formal House of Quality will be developed to connect sponsor/customer needs to measurable engineering characteristics. Based on the current project definition, likely customer needs include reliable attack detection, low false alarms, rapid response, minimal performance impact, compatibility with existing hardware, and practical prototyping cost.

Table 9.1: Customer Needs/Engineering Characteristics

Preliminary Customer Need	Related Engineering Characteristic	Current Target
Reliability	Attack detection rate	$\geq 95\%$
Avoid false alarms	False-positive rate	$\leq 5\%$
Responsive	Attack detection latency	≤ 1 s
Do not significantly slow host	RISC-V efficiency impact	$\leq 5\%$ reduction
Compatability	Current sensing range	500mA to 20A
Data Collection	ADC resolution	12 bits
Show normal operations	Reference waveform coverage	100% of test-plan references

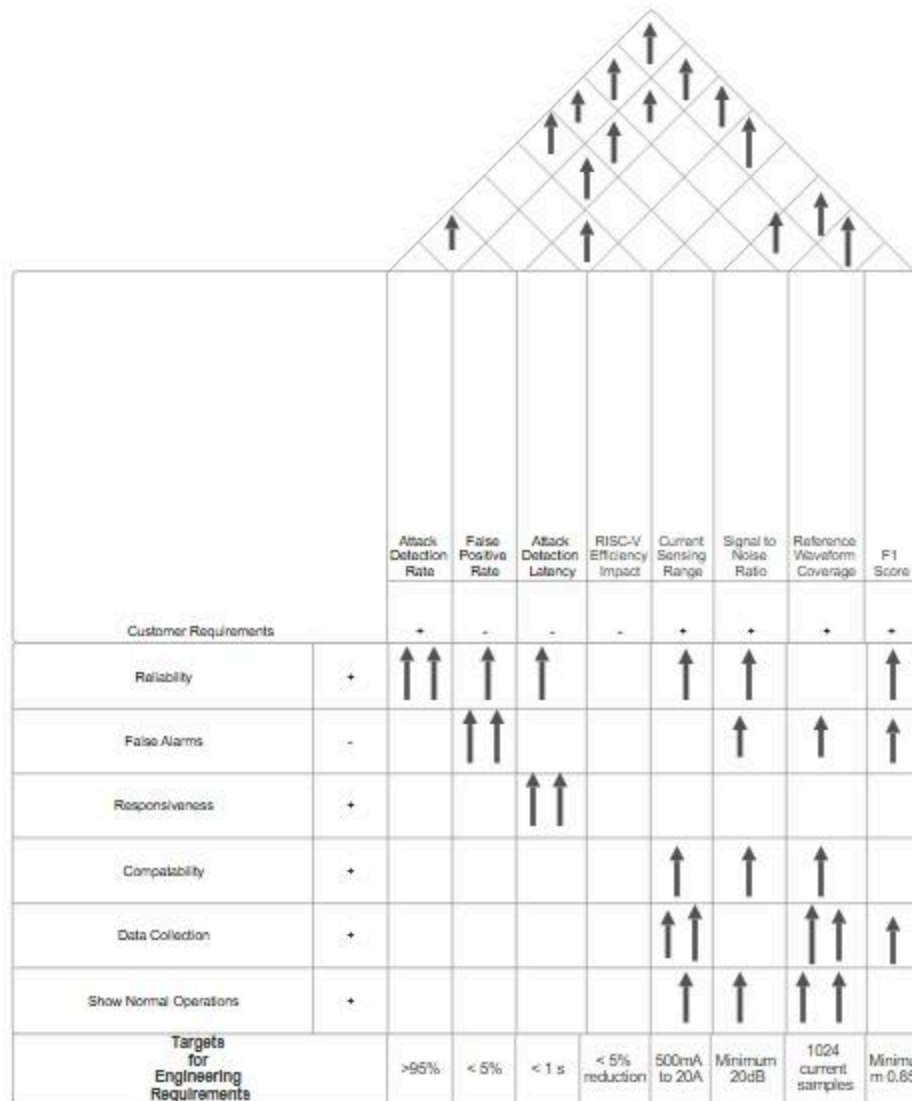


Figure 9.1 - House of Quality

10. Budget and Financing

NG PULSE is funded by Northrop Grumman with a maximum project budget of \$1,000. The team will use existing UCF Senior Design and DRACO laboratory equipment where available to reduce costs. The project bill of materials will be updated as exact components are selected.

Table 10.1: Cost Breakdown

Budget Item	Example Contents	Estimated Cost	Status
Current sensing	Sensor/amplifier, shunt, passives	[TBD]	To be selected
Data conversion	12-bit ADC and support components	[TBD]	To be selected
Detection/support	Comparator, logic/support ICs	[TBD]	To be selected
Power	Regulators, protection, filtering	[TBD]	Investigating
Prototype materials	Breadboard, wiring, connectors, headers	[TBD]	Planned
PCB/fabrication	Prototype PCB and assembly as required	[TBD]	Future
Contingency	Replacement parts/shipping/unplanned needs	[TBD]	Reserved
TOTAL	Must remain within sponsor funding	≤ \$1,000	Requirement

11. Project Milestones

11.1 Senior Design I

SD1 will focus on establishing the technical foundation needed to begin prototyping immediately in SD2. By the end of SD1, the team intends to have all major hardware selected, the golden profile obtained, a schematic of the proposed ASIC/system completed, and at least 80% of the Verilog code complete.

Table 11.1: SD1 Milestones

SD1 Milestone	Target	Completion Evidence
System definition and requirements	Early SD1	Approved architecture/specifications
Golden-profile acquisition	During SD1	Stored/identified reference waveforms
Hardware component selection	By end of SD1	Selected parts/BOM

ASIC/system schematic	By end of SD1	CAD schematic and connection review
Verilog development	≥80% complete by end of SD1	RTL modules/test results
Prototype readiness	End of SD1	Design ready to build at start of SD2

11.2 Senior Design II

Physical prototyping will begin immediately at the start of SD2. Initial work will focus on constructing the breadboard detector, integrating the FPGA and RISC-V target, and establishing basic end-to-end operation. The remaining approximately two months of SD2 will be devoted primarily to troubleshooting, refinement, final specification testing, data collection, and preparation for the final demonstration and presentation.

Table 11.2: SD2 Milestones

SD2 Milestone	Target	Completion Evidence
Begin physical prototype	Immediately at start of SD2	PCB sensing/detection path
Integrate digital logic/target	Early SD2	FPGA/RISC-V interface operating
Initial attack detection tests	Early-to-mid SD2	Attack flag observed
Troubleshooting and refinement	Final ~2 months	Resolved integration issues and improved reliability
Engineering-spec verification	Before final demo	Recorded pass/fail results for key specifications
Final presentation preparation	End of SD2	Demonstration-ready system and presentation materials

12. Team Organization and Work Distribution

Table 12.1: Work Distribution

Team Member	Primary Responsibility	Secondary Responsibility
Joey Esposito	ADC and supporting circuitry selection, comparator/protection circuitry, power distribution requirements and hardware component/BOM selection.	Hardware schematic development, FPGA/processor hardware interface, prototype assembly, electrical verification and specification testing
Dylan Ruiz	Current-sense design, current-sensor/amplifier selection, shunt/passive component selection, analog signal conditioning and power/voltage regulator design	Hardware schematic development, analog/ADC interface verification, breadboard prototype construction, hardware testing and construction
Sean Kenney	RISC-V/workload simulation, golden model and golden profile generation, windowed feature extraction RTL and Verilog module development	RTL testbenches, simulation/data analysis, ADC to digital logic integration, FPGA implementation and debugging
Jacob Daleandro	Threshold/classifier engine, attack detection and attack flag logic, PMBus/I ² C/status interface logic and Verilog module development	Golden-profile comparison logic, RTL verification, FPGA/RISC-V integration, digital-system testing and debugging

Both Dylan and Joey are electrical engineering majors and as such will select all hardware for this project together, including, but not limited to the current capture circuit, ADC, and voltage regulator. They also will be responsible for producing the schematic and verifying all power and connection requirements. Sean and Jacob are both computer engineering majors so they will be creating a virtual model of the RISC-V processor and obtain the golden profile from their simulation. They will write the verilog code to program the FPGA and Nordic board as well.

13. Technical Risks and Open Decisions

- The exact definition of the physical attack(s), including the high-current injection test method and safe laboratory limits, must be established with the sponsor/advisor.
- The exact golden-profile generation method, waveform feature set, and threshold/classifier algorithm remain to be finalized.
- The shutdown mechanism and interface to the protected system remain a stretch design decision.
- Exact component part numbers, sampling rate, sensor range/bandwidth, and detailed analog performance requirements remain to be selected.
- This project and its specifications are determined in large part by the team at Northrop Grumman and so they are subject to change depending on their direction.

14. Preliminary Verification and Test Strategy

Verification will proceed from golden-profile acquisition through virtual RTL verification and finally to breadboard prototype testing. Normal workloads will be used to establish the reference set and evaluate false-positive behavior, while defined attack trials will be used to evaluate detection rate, latency, F1 score, and shutdown functionality. Processor benchmarks will be run with and without NG PULSE attached to determine the impact on computing efficiency. Current-sensing performance and signal quality will also be evaluated to ensure that the acquired current waveforms are suitable for reliable anomaly detection.

- **Reference-waveform test:** For all four (4) selected processes, collect and average 1024 current samples to establish each golden profile. Verify that a golden waveform is successfully acquired for each selected process.
- **Efficiency test:** Run a defined RISC-V benchmark with and without NG PULSE attached. Pass if there is no reduction in computing efficiency (0% efficiency loss).
- **Latency test:** Measure the time from a detectable attack/anomaly to assertion of the alert flag. Pass if the detection latency is ≤ 1 s.
- **Detection-rate test:** Run repeated, defined attack trials. Pass if $\geq 95\%$ of attacks are correctly detected and flagged.
- **False-positive test:** Run repeated normal-workload trials. Pass if $\leq 5\%$ of normal trials are incorrectly identified as attacks.
- **Current-sensing range test:** Verify that the current-sensing system operates over the specified range of 500 mA to 20 A. Minimum-current detection will be verified within the specified accuracy, and a thermal imaging camera will be used under high-load conditions to verify safe operation.

- **Signal-to-noise ratio (SNR) test:** Compare acquired current waveforms to the established golden profile to evaluate signal quality relative to measurement noise. Pass if the measured SNR is ≥ 20 dB.
- **F1 score test:** Run repeated defined attack trials and normal-operation trials and calculate the F1 score from the resulting detection performance. Pass if the F1 score is ≥ 0.85 .
- **Long-time overcurrent shutdown test:** Apply a sustained current surge of 1.05–1.3 pu and verify that system shutdown occurs within 20 s.
- **Short-time overcurrent shutdown test:** Apply a current surge of 1.3–2 pu and verify that system shutdown occurs within 5 s.

15. Conclusion

NG PULSE proposes a hardware-security monitor that uses processor supply-rail current behavior as an indicator of workload integrity and abnormal physical activity. The project will establish golden current profiles for four selected processor processes, implement current sensing and digital detection hardware, verify the design virtually, and build a physical prototype for evaluation. The system will be evaluated using measurable performance criteria including detection latency, attack detection rate, false-positive rate, F1 score, signal-to-noise ratio, current-sensing range, processor efficiency impact, and overcurrent shutdown response.

The project is structured so that SD1 produces a prototype-ready design and SD2 begins immediately with integration and testing, followed by an extended troubleshooting and verification period. The current project definition provides measurable system targets and a clear development path, while several details remain intentionally open for team, sponsor, and advisor decisions. These details should be finalized as component selection, laboratory access, attack definitions, team assignments, and detailed design work progress.

Appendix A - References

Course guideline: EEL 4914 Senior Design I, Divide and Conquer Guideline. Additional technical references and product datasheets must be added as the prior-work review and component selection are completed.

[1] Texas Instruments “SBOS790 Data sheet | TI.com.”
<https://www.ti.com/document-viewer/INA233/datasheet>

[2] Analog Devices, Inc., “LTC2945: Wide Range I²C Power Monitor,” Rev. D.
<https://www.analog.com/media/en/technical-documentation/data-sheets/ltc2945.pdf>

[3] Analog Devices, Inc., “ADM1293/ADM1294: Digital Power Monitor with PMBus Interface,” Rev. B, 2015.
https://www.analog.com/media/en/technical-documentation/data-sheets/ADM1293_1294.pdf

Appendix B - Copyright Permissions

Appendix C - Supporting Material